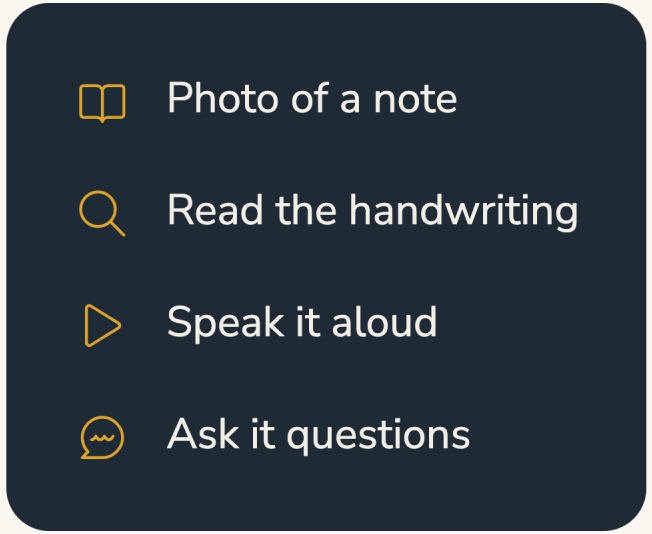
HANDS-ON WORKSHOP

Turn a handwritten note into an audiobook

Build a study buddy that reads your notes aloud and answers your questions about them, using free tools.

Yogen Ghodke · AI Solutions Engineer · builder of ListenToMe



WHY THIS MATTERS

Most of what students read never becomes an audiobook.

Class notes

Handwritten, photocopied, shared on WhatsApp. No publisher will ever record them.

Screen readers stop

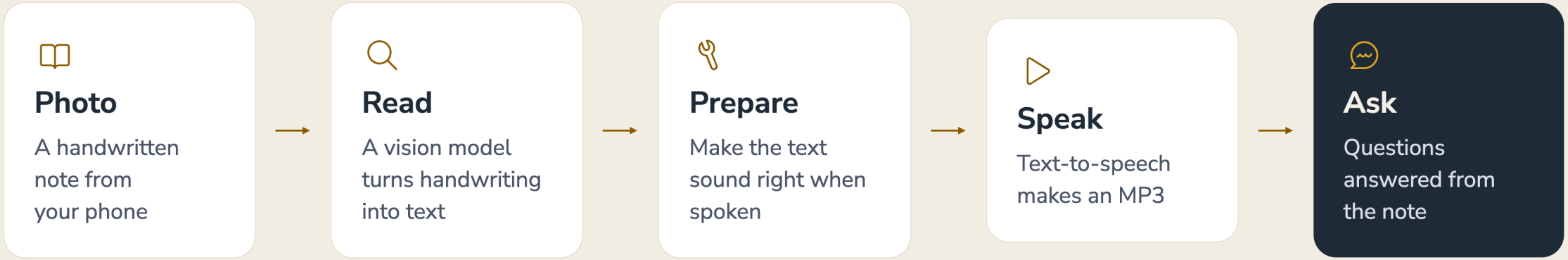
A photo of a page is just pixels to a screen reader. Blind students wait for someone to read it aloud.

Open knowledge

Aaron Swartz believed knowledge should be open to everyone. Today we make one page open to the ear.

WHAT WE BUILD

One page in, an audiobook and a tutor out



Four small steps, wrapped in a simple web app you vibe-code with Claude. Each step is useful on its own.

THE FREE TOOLKIT

Everything runs on free tools

Job	Tool	Cost
Write the code	Claude, free plan (vibe coding)	Free account
Read the handwriting	Gemini API, free tier	Free API key
Speak it	Kokoro, open-source text-to-speech	Free, runs on your laptop
Answer questions	Gemini API, free tier	Free API key
Glue	Python 3 and ffmpeg	Free

Models and free API key: aistudio.google.com/docs/models

Kokoro: github.com/hexgrad/kokoro

Kokoro's English voices are used here. Free tiers have rate limits; we handle them in step 4.

STEP 1 · READ

Turn the handwriting into text

```
from google import genai
client = genai.Client() # reads GEMINI_API_KEY
note = client.files.upload(file="note.jpg")
ask = "Transcribe this handwritten note exactly. Plain text only."
text = client.models.generate_content(model="gemini-2.5-flash", contents=[note, ask]).text
```

Ask for plain text. No markdown, no commentary, in reading order.

Never guess. Ask it to mark unreadable words as [unclear].

Check the start. Read the first lines yourself before you trust the rest.

STEP 2 · PREPARE

Write it for the ear

A voice reads exactly what you give it. Notes are full of symbols and shorthand that look fine and sound wrong.

Fix it with one more prompt: **“Rewrite this so it sounds natural read aloud. Spell out symbols and abbreviations. Keep every fact.”**

On the page	Read aloud
1st, 2nd	first, second
Ch. 3	Chapter three
A → B	A leads to B
e.g.	for example
~50%	about fifty percent

STEP 3 · SPEAK

Make the audiobook with Kokoro

```
# pip install kokoro soundfile
from kokoro import KPipeline
import numpy as np, soundfile as sf
tts = KPipeline(lang_code="a") # "a" = American English
parts = [audio for _, _, audio in tts(spoken, voice="af_heart")]
sf.write("note.wav", np.concatenate(parts), 24000)
# ffmpeg -i note.wav -b:a 96k note.mp3
```

Kokoro splits long text into sentences and speaks each one. You glue the pieces together, save, and you have an audiobook you can play on any phone.

STEP 4 · ASK

Ask AI: the study buddy part

```
question = "Explain the second point like I am new to it."
rules = "Answer only from this note. If it is not in the note, say so."
answer = client.models.generate_content(model="gemini-2.5-flash",
contents=[rules, text, question]).text
# speak the answer with the same Kokoro voice from step 3
```

Ground it. The note is the context, so answers stay on what the student actually wrote.

Quiz mode. “Ask me three questions about this note” turns revision into a conversation.

Pace it. Free tiers allow a few requests a minute: wait and retry, never hammer.

LIVE DEMO

A handwritten note becomes an audiobook

Photo from a phone → text → audio you can play anywhere → questions answered out loud.

Then your turn: vibe-code your own version with Claude's free plan.

TAKE IT FURTHER

Six ideas to keep hacking on

Whole books

PDFs with chapters and a table of contents.

Diagrams

Describe charts and figures in words.

More languages

Try other voices and test them with native speakers.

Quizzes

Spoken questions to revise before an exam.

Drag and drop

Drop a PDF or photo straight onto the page.

Accessible UI

Big buttons, clear labels, test with a screen reader.

THANK YOU

Every page deserves to be heard.

Questions now, or message me after the workshop.

Yogen Ghodke

AI Solutions Engineer, New York

App: listentomeapp.com

LinkedIn: in/yogenghodke

Kokoro: github.com/hexgrad/kokoro

Models: aistudio.google.com/docs/models